Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Bryer, Scott C Ph.D.		
Project Title:	Regulation of Macrophage Migration and Satellite Cell Proliferation during Skeletal Muscle Regeneration: Role of Urokinase Plasminogen Activator		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60608-1516	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	n/a		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Satellite cells (SC) appear necessary for skeletal muscle regeneration and recovery from atrophy. Urokinase plasminogen activator deficient (uPA-/-) mice display impaired muscle regeneration, potentially associated with decreased macrophage accumulation and SC proliferation. Hypothesis: uPA is required for macrophage migration and macrophage-induced proliferation of SCs following muscle injury. Aims: 1) determine whether accumulation of macrophages and proliferation of SCs following injury is impaired in uPA-/- mice, 2) determine whether macrophage and SC migration through a fibrin matrix is impaired in isolated uPA- /- cells and 3) determine whether macrophage induced proliferation of SCs is impaired in isolated uPA-/- cells. The goal of this study is to better understand the role of uPA and the function of macrophages and satellite cells in muscle remodeling, which may provide a target for therapeutic and medical interventions. Spaceflight is associated with muscle atrophy and return to weight bearing is associated with muscle injury. Urokinase plasminogen activator (uPA) is expressed by both macrophages and satellite		

	cells, both of which are important for muscle remodeling. Mice deficient in urokinase demonstrate severely imparied skeletal muscle regeneration associated with decreased accumulation of macrophages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Effective countermeasures against the atrophy and injury associated with spaceflight will require understanding of factors that regulate the function of cells involved in muscle remodeling. It is clear that the loss of uPA severely impairs the normal regenerative response of skeletal muscle as observed following immuno-histochemical analysis. Macrophages were labeled with an F4/80 antibody in regenerating fiber cross-sections and demonstrate impaired accumulation of F4/80 positive cells at all time points following injury (1, 3, 5 and 10 days). Satellite cells are required for skeletal muscle regeneration. BrdU is incorporated into the DNA of proliferating cells, including satellite cells. BrdU positive cells were visibly decreased 3-days post-injury in uPA deficient versus wild-type mice. This data supports a crucial role for uPA in satellite cell activation and/or proliferation.
Bibliography Type:	Description: (Last Updated: 09/09/2009)
Articles in Peer-reviewed Journals	Koh TJ, Bryer SC, Pucci AM, Sisson TH. "Mice deficient in plasminogen activator inhibitor-1 have improved skeletal muscle regeneration." Am J Physiol Cell Physiol. 2005 Jul;289(1):C217-23. Epub 2005 Feb 16. PubMed PMID: 15716324 , Jul-2005